Purpose and operating model

A two-player, private FH6 telemetry site for drivers who play at different times and want automatic, after-the-fact competition and analysis.

01 Default experience

01 DRIVE INDEPENDENTLY

Each driver plays Rivals, circuits, sprints or other events whenever convenient.

02 CAPTURE AUTOMATICALLY

Forza sends telemetry to an assigned VPS UDP port. No recorder button.

03 REVIEW LATER

The site groups attempts, compares accepted laps and shows driving differences.

02 First-release scope

AUTOMATED, WITHOUT MID-GAME ACTIONS

- Assign each player a **destination UDP port** on the VPS; source IP is only supporting evidence.
- Save original packets, capture timestamps and derived laps/runs for later reprocessing.
- Infer event type, route and car when evidence is available; preserve unknown attempts.
- Show lap records, rivalry history, aligned telemetry and corner-by-corner findings.

TWO DIFFERENT COMPARISON MODES

Ranked: only accepted complete laps/runs on a reliably matching route and under compatible rules. Maintain separate categories where necessary.

Exploratory: overlay Rivals, circuit, sprint or different-car recordings when the course or segment truly aligns. Clearly label differences; do not imply a formal win.

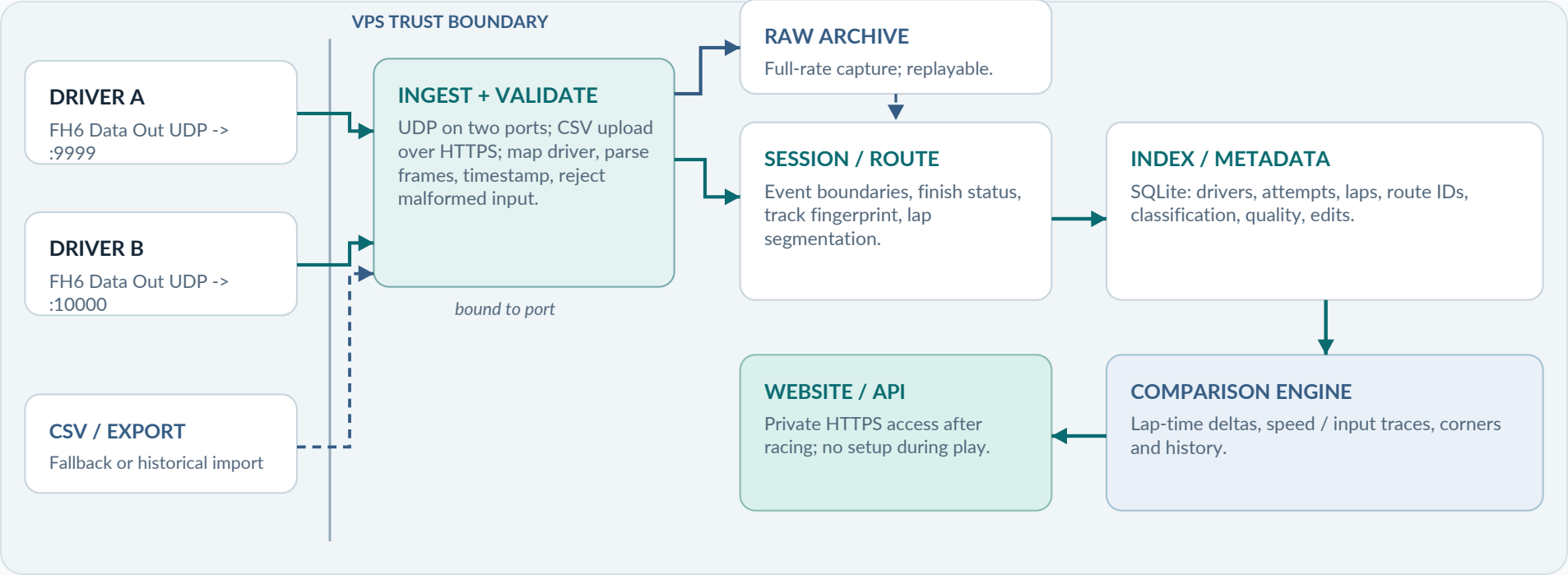
03 What is and is not known

STATUS	CURRENT UNDERSTANDING
Telemetry contains	Speed, throttle, brake, steering, world position, lap-related values and vehicle class/PI.
Can be inferred	Route identity, corners, event boundaries, partial/complete runs and suspected incidents.
Not directly transmitted	Official route name, game-approved lap validity, complete tune/assists, other players' car data.

Architecture and data flow

A deliberately small self-hosted system. Direct game UDP is sufficient for two streams; CSV remains a fallback and test input.

01 Proposed ingestion and analysis pipeline



02 Implementation and operational notes

RUNTIME

FastAPI / asyncio on the VPS, one UDP listener per player. SQLite (WAL) for indexed results and a simple HTTPS frontend.

LOAD + STORAGE

At 60 packets/s, 324 bytes each: ~70 MB raw per player-hour. One hour each = ~140 MB before indexes; low CPU/network load.

SECURITY

UDP has no built-in authentication or encryption. Port assignment identifies an input, but is not proof of identity. Limit ports, rate-limit, log, and protect site APIs.

Comparability, records and analysis

"Same map" is not necessarily "same race." Every result must retain route identity, event conditions and data-quality status.

01 Three keys to a meaningful comparison

ROUTE IDENTITY

Start and finish, direction, course geometry and measured path. A similar location alone does not prove the same route or checkpoints.

EVENT / TIMING RULES

Rivals, circuit races and point-to-point runs can use different timing conventions and conditions. Store their mode and classify matches.

CAR + CONDITIONS

Class, performance index and car ID are available. Tune/assists are not fully exposed; show known mismatches and optional annotations.

RANKED COMPARISON

Accepted complete laps/runs with matching course and agreed event/vehicle rules. Update PBs, track records and rivalry standings.

EXPLORATORY COMPARISON

Overlay speed, inputs, racing line and matching segments across modes. Differences are labelled, not necessarily fair timed races.

02 Data and analysis model

DATA ENTITIES + INDEPENDENT STATUS FIELDS

Driver -> **Capture** -> **Event attempt** -> **Lap / run** -> **Raw samples**.
A route, mode and car classification attach to each timed attempt. An individual capture can contain multiple events.

Track three separate dimensions: **completion** (completed / incomplete / provisional), **validity** (accepted / suspect / invalid), and **capture quality** (continuous / gapped / corrupt).

CORNER-LEVEL DIAGNOSIS

Align laps by **position along the same route**, then divide the course into stable corner segments. Compare brake onset, minimum speed, steering, throttle timing, exit speed and accumulated time delta.

Conclusions should expose the underlying traces. A lap delta shows **where** time changed; the suggested cause is an **inference**, especially when cars or conditions differ.

Important: completed laps may remain eligible even if the overall race was abandoned. A stream cut near the finish is not proof of a completed event.

Edge cases and expected system behaviour

Automate normal gameplay without inventing results when the telemetry is incomplete or ambiguous.

01 Behavioural rules (proposed for v1)

CASE	RECORD / CLASSIFY	SHOW / SCORE
Race quit after 2/3 laps	Keep the 2 completed laps; final lap marked incomplete; attempt DNF.	Completed accepted laps can rank; no overall race win.
Quit before any full lap	Keep partial raw samples and an incomplete attempt.	No lap time or record; shown in activity / diagnostics.
Rivals session ends	Close the capture; keep every completed lap.	Ending a Rivals session is **not** automatically a DNF.
Stream cuts near finish	Use multiple finish signals if available; otherwise provisional.	Do not assign a definite finish or record from ambiguous cutoff.
Internet loss / game crash	Store arriving frames; log gaps, pauses and last packet time.	Mark quality; suppress unsupported timing conclusions.
Restart / rewind / fast travel	Split attempts on restart; flag rewind; reject teleport as lap finish.	No spurious event merge or fastest lap.
Contact / dirty lap	Flag likely incidents; game does not send official invalidation.	Label suspect / user-reviewed, not game-verified clean.
Same roads, altered route	Different start/finish, direction or path => distinct route.	Separate full-lap records; optional shared-segment view later.
Different game types / cars	Keep event mode, car ID, class and PI with each attempt.	Rank separately by default; exploratory overlay where alignable.
Unknown custom route	Keep geometry under a temporary fingerprint; never guess its name.	Name/correct once and reclassify historical records.
Duplicate CSV / overlap	Use capture IDs, hashes and frame-time checks to deduplicate.	No duplicate laps or inflated competition statistics.
Changing IP / forged UDP	Bind driver to assigned destination port; source address is evidence only.	Log anomalies; consider VPN/collector if stronger trust needed.

Open decisions and definition of done

This is a proposed architecture, not a final implementation specification. The choices below should be made before formal scoring rules are fixed.

01 Still to be determined

- **What constitutes a rivalry win?** Best lap ever, weekly challenge, or individually matched attempts?
- **Ranking categories:** exact same car, class/PI only, or open/unrestricted?
- **Cross-mode:** which Rivals-versus-race results deserve a shared table versus an informational overlay?
- **Validity:** exclude inferred contacts / rewinds by default? Who can correct flags?
- **Route confidence:** how strict should automatic matching be, especially for custom routes?
- **Retention / privacy:** raw telemetry kept how long? Invite-only site, exports and deletion?
- **Transport:** direct public UDP acceptable initially, or is a VPN / local buffer worth adding?

DEFAULTS FOR A FIRST PROTOTYPE

- Keep **every attempted event**; exclude unfinished times from records.
- Use **destination UDP ports** for driver mapping, not source-IP/port alone.
- Use **separate ranked categories** for differing event rules, plus cross-mode visual comparisons.
- Treat guessed finishes and dirty laps as **suspect** until evidence or manual review.
- Store raw packets for reprocessing; launch with **SQLite** and a simple private HTTPS website.
- Avoid live-game actions, mandatory lap tagging and AI-generated claims without measurements.

02 Acceptance checks for the first working build

1. Each player configures FH6 Data Out once; unattended captures appear on the VPS on different days.
2. A and B remain distinguishable even if their public source IP is shared or changes.
3. Same recognized route matches across sessions; similar roads with different starts/directions do not merge.
4. Abandoned 3-lap event retains 2 completed laps; the open lap and race finish are not fabricated.
5. Rivals and conventional races on a shared route can be viewed together, with fair rankings separated.
6. One page displays PBs, clean/suspect labels, traces, largest corner time differences and capture quality.

RESEARCH BASIS FH6 Data Out | LapScope event detection / route matching | Forza-6-telemetry corner reports

Interface concept - after the session

A proposed detail-heavy view that both players can open later; these panels use invented data and are not a screenshot of a running system.



Expected features: inferred route and car, two-driver best times, geometric racing lines, speed / brake / throttle traces, corner differences, history and capture quality.